

Pseudo-evolution of the halo radius for different R_{vir} definitions

1 Purpose

The flow rates computed by SOAP are measured across a thin spherical shell of thickness ΔR placed at a radius R that tracks the halo boundary,

$$\dot{M}_{\text{flow}} = \frac{1}{\Delta R} \sum_{i \mid R-\Delta R/2 < r_i < R+\Delta R/2} m_i v_{r,i}, \quad (1)$$

where m_i is the mass of particle i and a positive velocity, and hence a positive flow rate, indicates outflow.

These flow rates include a correction for the pseudo-evolution of the halo radius: the radius of a static halo moves outwards because it is defined with respect to a reference density that decreases with time, so a particle at rest at the halo boundary is in fact inflowing. The correction subtracts $\dot{R}$ from the radial velocity of every particle,

$$v_{r,i} = (\mathbf{v}_i - \mathbf{v}_{\text{COM}}) \cdot \hat{\mathbf{r}} - \dot{R}, \quad \dot{R} \equiv \left(\frac{\partial R}{\partial t} \right)_M, \quad (2)$$

i.e. the velocity of the halo boundary at fixed enclosed mass.

SOAP computes flow rates for three spherical-overdensity (SO) definitions — 200c, 200m and BN98 — so $\dot{R}$ is needed for each of them. Section 2 derives a single expression that covers every SO definition, Section 3 collects the cosmological ingredients it requires, and Sections 4–6 evaluate it for the three cases in turn. The results are collected in Section 7.

2 General result

Every SO radius is defined by

$$M = \frac{4\pi}{3} \rho_{\text{ref}} R^3 \quad \implies \quad R = \left(\frac{3M}{4\pi \rho_{\text{ref}}} \right)^{1/3}, \quad (3)$$

where $\rho_{\text{ref}}(t)$ is the reference density appropriate to the definition. Taking the logarithm of Eq. (3) turns the cube root into a prefactor and the product into a sum,

$$\ln R = \frac{1}{3} \ln M - \frac{1}{3} \ln \rho_{\text{ref}} + \frac{1}{3} \ln \frac{3}{4\pi}, \quad (4)$$

and differentiating at fixed M removes both the mass and the constant,

$$\left(\frac{\partial \ln R}{\partial t} \right)_M = \frac{\dot{R}}{R} = -\frac{1}{3} \frac{\text{d} \ln \rho_{\text{ref}}}{\text{d} t}. \quad (5)$$

It is convenient to use $\ln a$ rather than t as the time variable. Since $\text{d} \ln a / \text{d} t = \dot{a}/a = H$, the chain rule gives, for any quantity $X(t)$,

$$\frac{\text{d} \ln X}{\text{d} t} = \frac{\text{d} \ln X}{\text{d} \ln a} \frac{\text{d} \ln a}{\text{d} t} = H \frac{\text{d} \ln X}{\text{d} \ln a}, \quad \text{equivalently} \quad \frac{\text{d} \ln X}{\text{d} \ln a} = \frac{\dot{X}}{HX}. \quad (6)$$

Applying this to ρ_{ref} in Eq. (5),

$$\frac{\dot{R}}{R} = -\frac{H}{3} \frac{d \ln \rho_{\text{ref}}}{d \ln a}, \quad (7)$$

so that

$$\boxed{\dot{R} = -\frac{RH}{3} \frac{d \ln \rho_{\text{ref}}}{d \ln a}} \quad (8)$$

The entire dependence on the halo definition is carried by the single dimensionless number $d \ln \rho_{\text{ref}}/d \ln a$. Note that Eq. (8) needs R , not M : using the SO radius that SOAP has already found avoids re-deriving it from the mass and makes the expression definition-agnostic. It also means that Eq. (8) applies directly to a shell placed at $R = fR_{\text{SO}}$, whose velocity is $f\dot{R}_{\text{SO}}$, provided the shell radius is the one substituted.

For completeness, a definition using a fixed proper radius falls outside Eq. (8) altogether: there is no reference density to differentiate, R is constant by construction, and $\dot{R} = 0$.

3 Cosmological ingredients

Two standard results are needed. First, differentiating $H = \dot{a}/a$ with respect to time,

$$\dot{H} = \frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2} = \frac{\ddot{a}}{a} - H^2, \quad (9)$$

and inserting the acceleration equation $\ddot{a}/a = -\frac{4\pi G}{3}(\rho + 3P/c^2)$,

$$\dot{H} = -\frac{4\pi G}{3} \left(\rho + \frac{3P}{c^2} \right) - H^2. \quad (10)$$

Eliminating H^2 with the Friedmann equation for a spatially flat universe, $H^2 = 8\pi G\rho/3$, the two contributions to ρ combine as $-\frac{4}{3} - \frac{8}{3} = -4$ and the factor 3 multiplying the pressure cancels against the prefactor,

$$\dot{H} = -\frac{4\pi G}{3}\rho - \frac{4\pi GP}{c^2} - \frac{8\pi G}{3}\rho = -4\pi G \left(\rho + \frac{P}{c^2} \right). \quad (11)$$

Dividing Eq. (11) by $H^2 = 8\pi G\rho/3$, and using Eq. (6) with $X = H$ to identify the left-hand side,

$$\frac{d \ln H}{d \ln a} = \frac{\dot{H}}{H^2} = -\frac{4\pi G(\rho + P/c^2)}{8\pi G\rho/3} = -\frac{3}{2} \left(1 + \frac{P}{\rho c^2} \right). \quad (12)$$

With $P/c^2 = \rho_r/3 - \rho_\Lambda$ and $\rho = \rho_r + \rho_m + \rho_\Lambda$ the dark energy terms cancel, $\rho + P/c^2 = \frac{4}{3}\rho_r + \rho_m$, and therefore

$$\frac{d \ln H}{d \ln a} = -\frac{3}{2} \left(\frac{4}{3}\Omega_r + \Omega_m \right) = -\left(2\Omega_r + \frac{3}{2}\Omega_m \right) \equiv -(1+q), \quad (13)$$

where q is the deceleration parameter and all density parameters are evaluated at the redshift of interest. Second, the critical density $\rho_c = 3H^2/8\pi G$ satisfies $\ln \rho_c = 2 \ln H + \text{const}$, so the exponent appears as a prefactor and

$$\frac{d \ln \rho_c}{d \ln a} = 2 \frac{d \ln H}{d \ln a} = -2(1+q). \quad (14)$$

It also follows, from $\Omega_m = 8\pi G\rho_m/3H^2 \propto a^{-3}/H^2$ and hence $\ln \Omega_m = -3 \ln a - 2 \ln H + \text{const}$, that

$$\frac{d \ln \Omega_m}{d \ln a} = -3 - 2 \frac{d \ln H}{d \ln a} = -3 + 2(1+q) = 4\Omega_r + 3\Omega_m - 3, \quad (15)$$

which correctly vanishes in the Einstein–de Sitter limit $\Omega_m \rightarrow 1$. This is needed for BN98 in Section 6.

4 Definitions with respect to the critical density

Here $\rho_{\text{ref}} = \Delta \rho_c$ with Δ a constant (200c, 500c, 2500c, ...). Since $\ln \rho_{\text{ref}} = \ln \Delta + \ln \rho_c$, the constant drops out of the logarithmic derivative and Eq. (14) gives $d \ln \rho_{\text{ref}} / d \ln a = -2(1 + q)$. Substituting into Eq. (8),

$$\dot{R}_{\Delta c} = -\frac{R_{\Delta c} H}{3} [-2(1 + q)] = \frac{2}{3} R_{\Delta c} H (1 + q), \quad (16)$$

that is,

$$\boxed{\dot{R}_{\Delta c} = \frac{2}{3} R_{\Delta c} H \left(2\Omega_r + \frac{3}{2} \Omega_m \right)} \quad (17)$$

This is independent of Δ : the same coefficient applies to 200c and 500c, provided the corresponding radius is used.

The radius itself follows from Eq. (3) with $\rho_{\text{ref}} = \Delta \cdot 3H^2 / 8\pi G$,

$$R_{\Delta c} = \left(\frac{3M \cdot 8\pi G}{4\pi \cdot 3\Delta H^2} \right)^{1/3} = \left(\frac{2GM}{\Delta H^2} \right)^{1/3}, \quad (18)$$

and substituting this closed form into Eq. (17), so that $R_{\Delta c} H = (2GMH^3 / \Delta H^2)^{1/3} = (2GMH / \Delta)^{1/3}$, gives the explicit expression in terms of the mass,

$$\dot{R}_{\Delta c} = \frac{2}{3} \left(\frac{2GMH}{\Delta} \right)^{1/3} \left(2\Omega_r + \frac{3}{2} \Omega_m \right) \xrightarrow{\Delta=200} \frac{2}{3} \left(\frac{GMH}{100} \right)^{1/3} \left(2\Omega_r + \frac{3}{2} \Omega_m \right). \quad (19)$$

The coefficient 100 is $\Delta/2$ and is therefore specific to 200c; it would be 250 for 500c and 1250 for 2500c. Note that this substitution is the only place where the choice $\Delta = 200$ enters — Eq. (17) itself is general. Using Eq. (17) with the known radius therefore removes the hard-coded constant altogether.

5 Definitions with respect to the mean matter density

Here $\rho_{\text{ref}} = \Delta_m \rho_m \propto a^{-3}$, so $\ln \rho_{\text{ref}} = \text{const} - 3 \ln a$ and $d \ln \rho_{\text{ref}} / d \ln a = -3$ exactly. Eq. (8) then collapses to

$$\dot{R}_{\Delta m} = -\frac{R_{\Delta m} H}{3} (-3) \quad \Rightarrow \quad \boxed{\dot{R}_{\Delta m} = R_{\Delta m} H} \quad (20)$$

independently of Δ_m and of cosmology. This is simply the statement that at fixed mass a radius defined relative to the mean matter density is constant in comoving units, so its proper velocity is pure Hubble flow. Note that this is not a small correction to Eq. (17): at $z = 0$ it is larger, for the same R , by a factor $3/[2(1 + q)] \simeq 3.3$, while the two agree in the matter-dominated limit $1 + q \rightarrow 3/2$, where $\rho_c \propto \rho_m$.

6 The BN98 definition

Bryan & Norman (1998) define the halo with respect to the critical density but with a time-dependent contrast,

$$\rho_{\text{ref}} = \Delta_{\text{BN}}(a) \rho_c(a), \quad \Delta_{\text{BN}} = 18\pi^2 + 82x - 39x^2, \quad x \equiv \Omega_m(z) - 1, \quad (21)$$

the last equality holding for a flat universe.¹ Both factors now evolve. Taking the logarithm of Eq. (21), $\ln \rho_{\text{ref}} = \ln \Delta_{\text{BN}} + \ln \rho_c$, the contrast no longer drops out and Eq. (14) supplies only the second term,

$$\frac{d \ln \rho_{\text{ref}}}{d \ln a} = \frac{d \ln \Delta_{\text{BN}}}{d \ln a} - 2(1 + q), \quad (22)$$

so that Eq. (8) gives

$$\boxed{\dot{R}_{\text{BN98}} = \frac{R_{\text{BN98}} H}{3} \left[2 \left(2\Omega_r + \frac{3}{2}\Omega_m \right) - \frac{d \ln \Delta_{\text{BN}}}{d \ln a} \right]} \quad (23)$$

The remaining derivative follows from Eqs. (21) and (15) by the chain rule. Since Δ_{BN} depends on a only through x , and $dx/d \ln a = d\Omega_m/d \ln a$,

$$\frac{d \ln \Delta_{\text{BN}}}{d \ln a} = \frac{1}{\Delta_{\text{BN}}} \frac{d\Delta_{\text{BN}}}{dx} \frac{dx}{d \ln a} = \frac{(82 - 78x)}{18\pi^2 + 82x - 39x^2} \Omega_m (4\Omega_r + 3\Omega_m - 3), \quad (24)$$

so the result is fully analytic. As required, Eq. (24) vanishes for $\Omega_m \rightarrow 1$ and Eq. (23) then reduces to Eq. (17), which in turn agrees with Eq. (20) in that limit.

7 Summary

Writing $\dot{R} = cRH$, the coefficient c for each definition is

Definition	ρ_{ref}	$d \ln \rho_{\text{ref}}/d \ln a$	c
Δ_m (e.g. 200m)	$\Delta_m \rho_m$	-3	1
Δ_c (e.g. 200c)	$\Delta_c \rho_c$	$-2(1 + q)$	$\frac{2}{3}(1 + q)$
BN98	$\Delta_{\text{BN}}(a)\rho_c$	$d \ln \Delta_{\text{BN}}/d \ln a - 2(1 + q)$	$\frac{1}{3} [2(1 + q) - d \ln \Delta_{\text{BN}}/d \ln a]$
fixed proper radius	—	—	0

with $1 + q = 2\Omega_r + \frac{3}{2}\Omega_m$ and $d \ln \Delta_{\text{BN}}/d \ln a$ given by Eq. (24). Evaluating these for a Planck-like cosmology ($\Omega_m^0 = 0.3046$, $\Omega_\Lambda = 0.6939$, $\Omega_r^0 = 7.79 \times 10^{-5}$, $\Omega_\nu^0 = 1.389 \times 10^{-3}$):

z	0	0.3	1	2	3	5
$\Omega_m(z)$	0.306	0.492	0.779	0.922	0.965	0.988
$\Delta_{\text{BN}}(z)$	102.0	125.9	157.6	171.0	174.7	176.7
c (200m)	1.000	1.000	1.000	1.000	1.000	1.000
c (200c)	0.306	0.492	0.779	0.923	0.966	0.990
c (BN98)	0.590	0.733	0.888	0.959	0.982	0.995

All three converge at high redshift, as they must: in the Einstein–de Sitter limit every definition comoves with the expansion and $\dot{R} \rightarrow RH$.

¹SOAP evaluates $x = -(\Omega_k a^{-2} + \Omega_\Lambda) / (\Omega_k a^{-2} + \Omega_m a^{-3} + \Omega_\Lambda)$, which reduces to $\Omega_m(z) - 1$ for $\Omega_k = 0$ and neglected radiation.